

GEO-ANOM Task 1 — Field Labeling Guide

Decision rules, the specific farms worth checking in person (with the **real structure coordinates**), and the satellite screenshots behind every call. Updated 2026-09-23. Companion file: GEO-ANOM_Task1_Field_Map.kmz (same layers, offline in Google Earth).

What changed in this version — read first

- **Registry permit points are often hundreds of meters off.** Most beef/dairy farms checked were 400-860 m from their real barns (Brandenburg 861 m, Bistate 845 m, Lester Jones 618 m, Oakland View 615 m, Panorama 423 m; Horizon was close at 94 m) and two dairy permits do not sit on a dairy at all. Navigate to the structure coordinates in §4, not the permit pin.
- **Detection outlines are often partial.** Brandenburg's outline covers about half its barn; Panorama's covers part of one barn in a complex of several. Floor-area numbers for non-poultry farms are underestimates.
- **Lagoon records were checked against imagery and mostly do not hold up.** Of the 7 hand-recorded lagoons only Dulin looks real. Tran and Roland Todd (recorded 'confirmed') are roofs / not visible; Rahim is a house pond; Hutchison is crop field. The most common false positive is a blue-gray metal roof.
- **Two dairy farms are true model misses** (Lester Jones, Horizon: big conventional barns + lagoons at the location, zero output), **two are registry errors** (Fry, Oak Bluff).

How to read the pictures

-  Building detection kept by the pipeline (label = length x width)
-  Raw model candidate that the shape filter dropped
-  Hand-recorded lagoon polygon ('confirmed' in the file)
-  Hand-recorded lagoon polygon, 'uncertain'
-  Automated lagoon candidate
-  Automated candidate marked 'CAFOSat-corroborated'
-  Known false positive
- + White crosshair = MDE registry permit point (geocoded address; may be far from the barns)
- o White ring + label = analyst annotation eyeballed from the imagery (+/- ~30 m), not model output

Imagery: 2023 NAIP (Planetary Computer, ~1 m/px) unless a figure says MD iMAP. Left panel of each pair is clean imagery — judge it before looking at the boxes.

1. What We're Tracking Right Now

Species	Farm-level coverage	Status (with today's caveats)
Poultry	358 / 389 farms	Validated: $R^2=0.655$ vs. ground truth, 99.7% precision, 0.9% true-miss rate
Swine	1 / 1 farms	Detected -- one barn outlined
Beef	3 / 3 farms	Detected, but outlines are partial and 2 came from a threshold exception on 2026-09-21; both now confirmed to be real barns on imagery (§5 A1-A2)
Dairy	1 / 14 farms	Open. Lester Jones + Horizon: real barns and lagoons, zero model output. Fry + Oak Bluff: permit point is not on a dairy. Other 10: not yet reviewed.
Lagoons	1 of 7 recorded looks real	Automated list ~10% precise; dominant error = blue metal roofs read as water

2. Decision Rules — Buildings

Structure seen on the ground	Call	Why / evidence
Broiler house (long, enclosed gray/white roof)	YES	The pipeline's home case -- §5 C1
Layer / turkey / duck house	YES	Same signature; layers 7/7 on Cal-Maine (C3), turkeys (C4), ducks (C5)

Structure seen on the ground	Call	Why / evidence
Swine confinement barn	YES	Same signature (Grand View, C2)
Beef/heifer confinement barn	YES, outline partial	Bistate is fully outlined (A3); Brandenburg/Panora only partly (A1, A2)
Dairy free-stall barn	MOSTLY MISSED	Conventional long-roof barns at Lester Jones and Horizon returned nothing (A4, A5); Oakland View gets a partial outline (A8)
Round grain/feed silo	NO (correct)	Circular shape fails the aspect-ratio floor
Grandstand / stable / warehouse (non-farm)	NO -- false positive	Pimlico Race Course, 4 detections (D1); removed from Task 1 output

3. Decision Rules — Water Bodies (Lagoons)

What the imagery shows	Call	Example / note
Regular banked pond, teal/brown/dark, next to barns	YES	Dulin (B1); Lester Jones's three lagoons and Horizon's three ponds (A4, A5)
Blue-gray metal roof (barn or shed)	NO -- #1 false positive	Tran (B2), Wolf Farm (D2), Moore (D3), Hite/Nguyen (B5): roof color reads as water
Pond beside a house / pool	NO	Rahim (B4) -- not the 'river fragment' the older guide says
Ordinary crop field	NO	Hutchison's three candidates (A9)
ROUND concrete manure tank (dark liquid)	REAL, currently invisible	Two at Panora (A2). Lagoon-type storage the rectangular-pond detector is not built to find
Natural pond in a forest clearing; quarry/borrow pit; river fragment	NO	Documented in docs/labeling_guide.md; none of these is pictured here
Stormwater retention pond on a farm	UNRESOLVED	Flagged as a real confusion class, no rule yet -- if you see one, note its position relative to barns

The lagoon step keeps only candidates near detected barns, so a farm whose barns are missed (Lester Jones, Horizon) also loses its lagoons.

4. Field Priority — Where To Go and What To Check

Coordinates are lat, lon of the **actual structures** (detected or eyeballed), followed by how far the registry permit point is. Screenshot numbers refer to §5.

4a. Beef — are the two added detections real barns?

Farm	County	Go to (structure)	Structure is, from permit pt	What to check
Dwight Brandenburg / Brandenburg Family LP	Frederick	39.40455, -77.62259	861 m SE	Imagery shows a real ~65x27 m barn, silage bags, a round manure store (A1). Species? Head count? What is the round tank?
Panora Acres / Stacy Sellers	Carroll	39.69321, -76.89273	423 m W	Large complex: several barns, silos, two round manure tanks (A2). Beef or dairy? How many barns? Tank contents?
Bistate Feeders (reference)	Caroline	38.97296, -75.77182	845 m N	Already fully detected (A3). Long narrow barns 111-153 m -- unlike the two above.

4b. Dairy — model misses vs. registry errors

Farm	County	Go to	Structure is, from permit pt	What to check
Lester C. Jones & Sons (3,300 head)	Kent	39.32006, -75.81566 (eyeballed)	618 m W	MODEL MISS. 4 large long-roof barns + 3 lagoons (A4). Roof type/material? Barn width? Is it open-sided or ridge-vented?
Horizon Organic Dairy (544 head)	Kent	39.29126, -76.00813 (eyeballed)	94 m E	MODEL MISS. Cross-shaped light-roof barn + 3 ponds at the permit location (A5). Same questions.
Matthew Fry / Fair Hill (1,050 head)	Kent	39.25686, -76.06876 (eyeballed, tile edge)	982 m W	REGISTRY ERROR? Permit point is at an empty road junction (A6). Where is the real farm address?
Oak Bluff Dairy Farms (1,280 head)	Frederick	39.52347, -77.29351 (permit pt)	--	REGISTRY ERROR? No dairy visible near the point; large quarry to the west (A7). Where is the farm?
Oakland View Farms (reference)	Caroline	39.01206, -75.86124	615 m NE	The one dairy with a kept detection -- two long barns, partial outline (A8). Compare its roof with Lester Jones's.

The other 9 dairy permits (Teabow, Mason Dixon, Coldsprings, Whitelyn, Cow Comfort Inn, Patterson, My Lady's Manor, Deerspring, Arbaugh's) have not been checked on imagery yet.

4c. Lagoons — recorded records worth a real look

Record	County	Go to	What to check
Roland Todd -- 'confirmed'	Dorchester	38.68623, -75.84334	Imagery sources disagree by several hundred meters and neither shows a lagoon (B3). Is there a lagoon on this farm? Where?
Hoa Tran -- 'confirmed'	Wicomico	38.29693, -75.53077	Polygon sits on a poultry-house roof (B2). Any lagoon anywhere on this farm?
Paul Aaron Hutchison Sr. (3 candidates)	Talbot	38.80485, -76.01697	Crop field, no water visible (A9). Any pond here at all?
Dulin -- 'confirmed' (reference)	Caroline	see KMZ	The one that looks right (B1): what does a real one look like from the ground?

Hand-recorded lagoons were registered on older MD iMAP imagery: in Google Earth their position can be off by up to ~100 m.

4d. Registry entries with no animal type

All nine have detected buildings and every one looks like a broiler farm (\$5 E1). If you can confirm what a farm raises, that is a direct fix to the registry.

Farm	County	Permit point (lat, lon)	Buildings
Dustin Calloway/Clay Island Farm	Wicomico	38.42979, -75.78840	19
Bawi Hlun/ Chan UK	Somerset	38.25310, -75.64924	9
Hannah Jones	Caroline	38.90793, -75.77524	8
Adam Stanley/Deathly Hollows	Somerset	38.02565, -75.80555	7
Eldwin Martin/Airview Farm	Wicomico	38.36011, -75.52343	7
Robert Rosado/Northwind Farm	Wicomico	38.43855, -75.62340	4
Muhammad Usman/H & N Farm, LLC	Wicomico	38.35939, -75.32408	4
Chasin Dreams Farm	Queen Anne's	39.12587, -75.90554	2
Alan & Kristin Hudson	Worcester	38.36420, -75.25488	1

5. Satellite Screenshots

A. Field priority

[A1] Beef: Dwight Brandenburg / Brandenburg Family LP (Frederick)



Detection added 2026-09-21 by the beef-specific shape exception. A real barn (~65 x 27 m) with silage bags and a round manure store -- but the green outline covers only about half of the roof. The barn is 861 m SE of the registry point; the permit point falls in a hay field, so the registry geocode is off for this farm. Left: clean imagery. Right: green = pipeline detection.

[A2] Beef: Panora Acres / Stacy Sellers (Carroll)



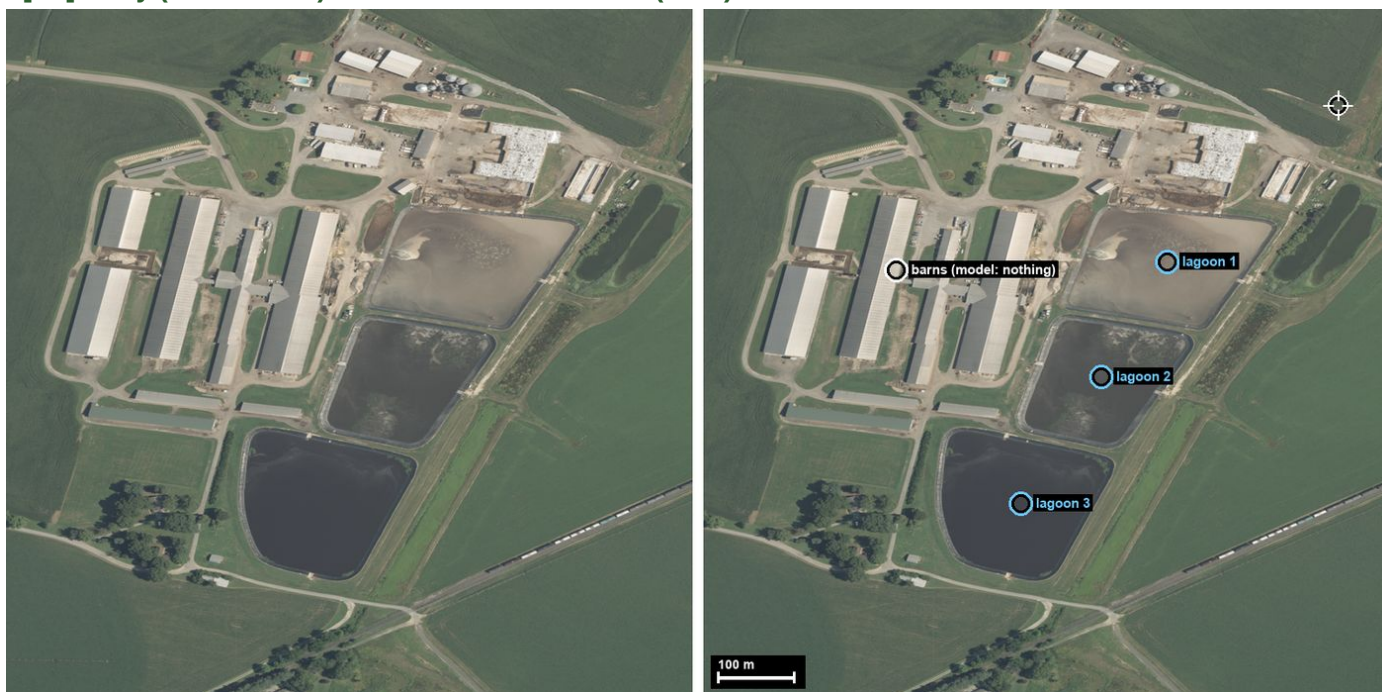
Second beef detection added 2026-09-21. This is a full livestock complex -- several large barns, feed silos, and TWO ROUND MANURE STORAGE TANKS (dark liquid). The pipeline outlined part of just one barn; the round tanks are a lagoon-type structure the lagoon detector is not built to find. The barn is 423 m W of the registry point.

[A3] Beef reference: Bistate Feeders (Caroline) -- the beef farm that already worked



7 detections, all long narrow barns (111-153 m x 12-14 m). The Brandenburg/Panora barns are shorter and wider than these.

[A4] Dairy (model miss): Lester C. Jones & Sons (Kent)



Large conventional long-roof dairy barns and three manure lagoons sit ~620 m WSW of the permit point (crosshair, top right). The model returned ZERO candidates for this whole 2 km tile. Labels are analyst annotations from the imagery (+/- ~30 m), not model output.

[A5] Dairy (model miss): Horizon Organic Dairy (Kent)



A cross-shaped long barn with a light roof, three manure ponds, and other outbuildings right at the permit location; the model returned ZERO candidates. Analyst annotations, +/- ~30 m.

[A6] Dairy (geocode error): Matthew Fry / Fair Hill Farms (Kent)



Whole 2 km tile. The permit point (crosshair) is at an empty road junction; the only dairy-looking complex (large blue-roof barn + dark lagoon) is ~1 km WEST at the tile edge. Likely a registry geocoding error, not a model failure -- the real farm may extend outside this tile.

[A7] Dairy (geocode error): Oak Bluff Dairy Farms (Frederick)



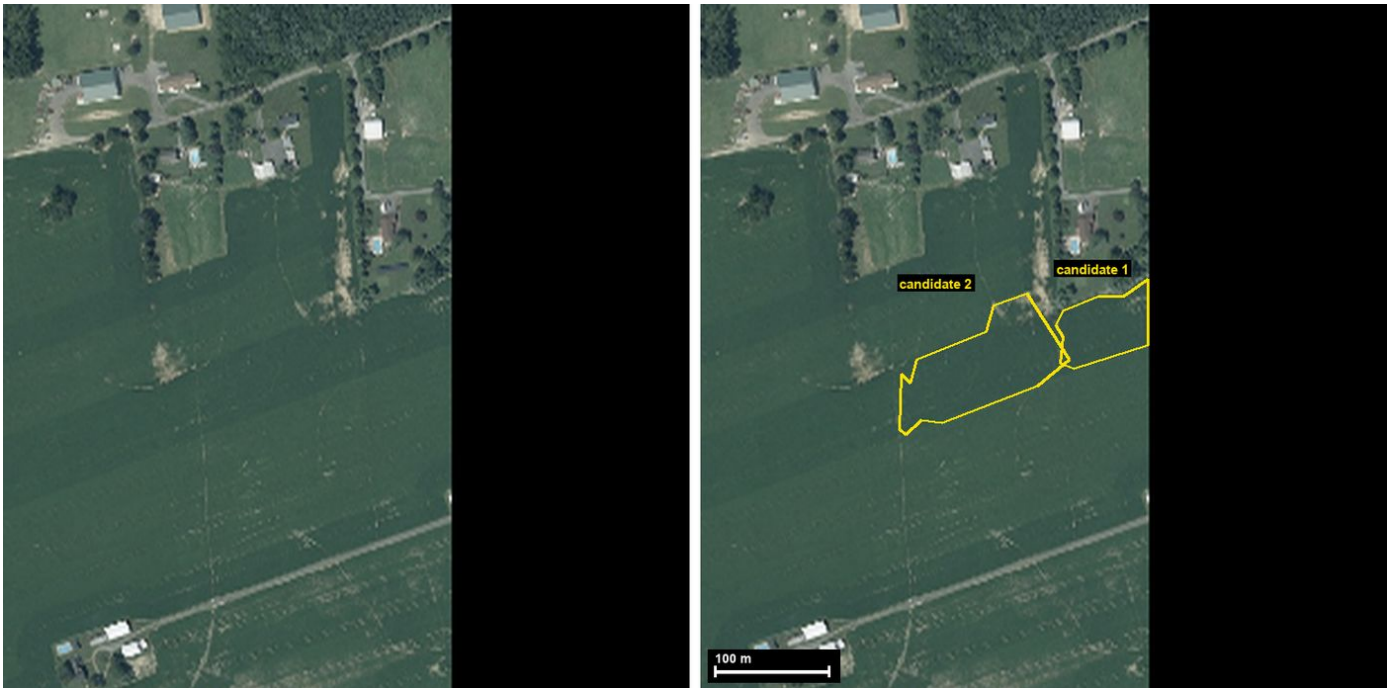
Whole 2 km tile. No dairy operation is visible near the permit point (crosshair): wooded land, a large quarry to the west, scattered houses. Likely a registry geocoding error; the farm is probably elsewhere.

[A8] Dairy reference: Oakland View Farms (Caroline) -- the one dairy that works



1 detection (615 m NE of the registry point). Two long barns are visible; the green outline covers only about half of one and orange fragments (dropped by the filter) mark parts of the other. This is the one dairy where the pipeline produced a kept detection -- and even here the outline is partial.

[A9] Lagoon candidates: Paul Aaron Hutchison Sr. / Home Farm (Talbot)



Recorded status 'uncertain' (3 candidates; #3 is ~400 m north, off this crop). On the imagery the polygons sit on ordinary crop field -- no water, no structure. Likely false positives; a look on the ground settles it. Yellow = uncertain. Black band = edge of the tile.

B. The recorded lagoons vs. the imagery

[B1] Dulin (Caroline): recorded 'confirmed' -- looks like a real lagoon



A clear teal, regular, banked pond ~380 m from the barns. Polygon aligns on both imagery sources. This is the one recorded lagoon that looks right.

[B2] Tran (Wicomico): recorded 'confirmed' -- polygon is on a poultry house roof



The recorded lagoon polygon outlines a gray-blue metal roof with ridge lines -- a poultry house, not water. The same on both imagery sources (black band = tile edge).

[B3] Roland Todd (Dorchester): recorded 'confirmed' -- imagery sources disagree



Left: MD iMAP imagery -- the polygon outlines three poultry houses, not water. Right: Planetary Computer -- the same coordinates are wood edge/field with no barns. The two sources disagree about what is at this spot by several hundred meters, and neither shows a lagoon inside the polygon. Status unverified -- the recorded 'confirmed' label is not supported by the imagery.

[B4] Rahim (Dorchester): recorded 'false positive' -- a pond beside a farmhouse



Small pond next to a house and pool, ~500 m from the barns, near a large river. Not a river fragment as the labeling guide says, but also not lagoon-like by siting. Treat as false positive unless the ground says otherwise.

[B5] Hite / Nguyen (Worcester): 'CAFOSat-corroborated' candidates



Left (south cluster): one polygon on a small white-roofed building at a forest edge, one on a lawn patch. Right (north cluster, ~850 m away): one on the end of a poultry-house roof, one on a small building. None of the four is water. The CAFOSat match was a loose 300 m patch-level check and does not confirm these specific polygons.

C. What correct detections look like

[C1] Poultry (broiler) reference: Alan Butler / A&P; Farm



14 detections; the pipeline outlines each long broiler house.

[C2] Swine reference: Grand View Farm (Kent)



Swine confinement barn -- same long enclosed roof shape as a poultry house. 1 detection.

[C3] Layers reference: Cal-Maine Foods, Cecil County



7 detections on a layer complex.

[C4] Turkeys reference: Jon Sewell / Tuscarora Farms (Frederick)



3 detections.

[C5] Ducks reference: Isaac Martin (Washington)



1 detection; the sister duck farm (Eldwin D. Martin) has none.

D. False positives

[D1] False positive: Pimlico Race Course grandstand/stable roofs



4 'poultry house' detections on a horse-racing venue (roofs share the long-roof shape). Excluded from the Task 1 output. Green = what the pipeline drew.

[D2] False positive: blue-gray metal barn roofs read as 'water' (Wolf Farm)



Purple = automated lagoon candidates. They outline the blue-gray metal roofs of poultry houses -- the roof color reads as water to the color-based candidate step. This is the most common lagoon false positive seen (also Tran, Roland Todd, Moore).

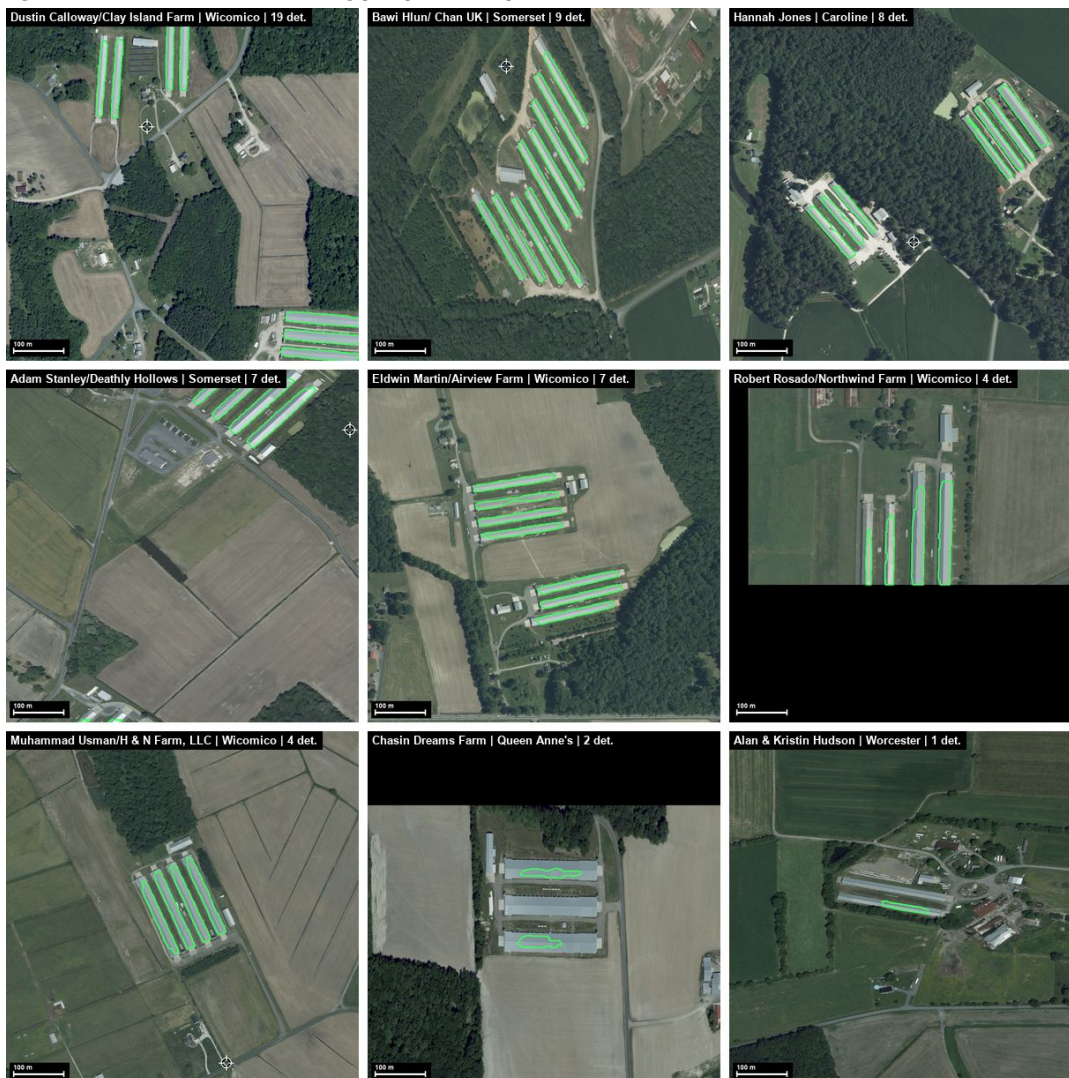
[D3] False positive: blue metal roofs again (William Moore, III)



Purple = automated lagoon candidates sitting on two blue metal-roofed buildings. (Forest-shadow-strip false positives are documented in the labeling guide but none is shown here.)

E. Registry entries with no animal type

[E1] Registry entries with no animal type (9 farms)



All nine have detected buildings (green) but no species/headcount on file. If you can tell what a farm raises, that is a direct fix to the registry.

GEO-ANOM Task 1 Field Labeling Guide, 2026-09-23. Screenshots regenerate with scripts/render_field_screenshots.py; the KMZ with scripts/build_field_kmz.py; this PDF with scripts/build_field_guide_pdf.py. Technical background: docs/labeling_guide.md, docs/research_log.md.